

Hennovate the State

CourtBot5000

Ask less. Find more.

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PROBLEM

- Delaware court website is large and complex to navigate
- Information spread across pages, forms, and courts
- Users need the right keyword to find the right page
- Overlapping content makes navigation confusing
- Added friction for mobile and multilingual users
- Maintenance grows harder as content grows

Our goal: make Delaware Courts information easier to find without changing the underlying court system.

SOLUTION

CourtBot5000 is a plain-language assistant that answers questions using only official Delaware Courts content and points users to the exact page or form they need.



What's New or Creative

- Retrieval locked to an allowlist of official sources — no open-web answers
- Hybrid lexical + semantic search handles terms users can't name or spell
- Every answer links back to the official page or form for verification
- PII-safe query logs surface content gaps for court staff
- Internal Audit Tool – Monitors website health, detects content issues, and helps court staff prioritize fixes using analytics and AI.

WHAT WE BUILT

CourtBot5000 Architecture

Solution architecture | Public navigation and employee website audit

AWS + Azure
Separate trust boundaries

PUBLIC RESIDENT EXPERIENCE

HTTPS | Same-origin web application



EMPLOYEE AUDIT EXPERIENCE

Separate service | Authenticated deployment



● AWS runtime and AI

● Azure document audit

● Protected storage path

No cloud credentials or raw analytics in browser code

RESULTS

Two Experiences, One Improvement Loop

Public court navigation for residents | Operational visibility for court employees

1. Public CourtBot5000

 **CourtBot5000**
Official website navigation assistant

RESIDENT QUESTION

Where can I find Family Court forms?

COURTBOT5000

Family Court forms are available on the official Delaware Courts website.

I can help you open the relevant forms and instructions.

OFFICIAL SOURCES ONLY

OFFICIAL COURT PAGES

 Family Court Forms
courts.delaware.gov

Open

 Family Court Instructions
courts.delaware.gov

Open

SANITIZED
AGGREGATES ONLY

2. Employee Audit Workspace

 **Delaware Courts**
EMPLOYEE ANALYTICS

QUERY ANALYTICS | WEBSITE HEALTH | CONTENT QUALITY | DOCUMENT AUDIT | RESOURCE FINDER

TOTAL EVENTS

840

Selected period

RETRIEVAL RATE

76%

ACTIVE FINDINGS

108

Across content checks

Top query topics

Forms 

Traffic 

Locations 

Content severity

 **108**

■ High 3
■ Medium 54
■ Low 51

Review queue

☐ Potential information mismatch

3 PAGES

REVIEW

AuditBot5000

Resident outcome: reach the correct official court resource

Staff outcome: find, review, assign, and track website issues

IMPACT FOR DELAWARE



Simpler Access

Ask in plain language instead of hunting through pages.



Better Navigation

Points users to official pages, forms, and courts.



Access to Justice

Lowers the navigation barrier to public information.



Efficiency

Potential to reduce repeated navigation questions.



Staff Visibility

Surfaces common questions and content gaps.



Increased Trust

Responses stay tied to official court resources.

RESPONSIBLE AI & DATA

Data & Scope

- Public Delaware Courts web content only
- No legal advice
- No live or private case database access
- No confidential court records used
- Grounded in available official content

Safety, Privacy & Access

- PII-aware query sanitization
- Sanitized retention for approved analytics
- Input and output safety controls
- Accessibility-focused, mobile-friendly design
- Multilingual support considerations
- Human review remains important for deployment

COMPUTE PLATFORM & LEARNING

GCP

Website scraping and
data acquisition

AWS

Bedrock generation
S3 query analytics
App hosting

Azure

Functions
Table & Blob Storage
Data analytics

Hybrid Retrieval Pipeline



The LLM is one component — retrieval, source control, privacy, safety, and evaluation matter just as much.

PROTOTYPE → PRODUCTION

Prototype Today

- Public Delaware Courts assistant
- Official-source retrieval, hybrid search
- Bedrock generation with guardrails
- Mobile-responsive interface
- Internal audit prototype
- Resource Finder on the public corpus
- Privacy-preserving query analytics

Production Path

- Formal Delaware Courts security review
- Agency pilot
- Authentication / role-based access
- Approved internal source integration
- Human review and governance process
- Continuous monitoring, red-team testing
- Accessibility and multilingual checks
- Content-update pipeline and evaluation

Takeaways

Mentors

On the Day 1: cautious about using AI in a court setting.

Today: more confident that AI, used responsibly, can help the public reach official court information.

The hope: technology that supports access to justice — with human oversight.

Students

- Hands-on experience on a real Delaware Courts problem
- Deeper understanding of responsible AI, accuracy, and public-sector constraints
- Direct collaboration with state and industry mentors

CourtBot5000 does not simplify the courts — it simplifies the experience of finding the right official information.